

Experiment 3: Survey Report on Emerging Trends in Low-Code Application Development

1. AIM

To study the emerging trends in low-code application development and analyse how modern low-code platforms reduce the need for traditional programming while enabling faster design, testing, deployment and maintenance of applications.

2. Traditional Application Development vs Emerging Trends in Low-Code Application Development

Traditional Application Development	Low-Code Application Development
Requires extensive programming knowledge.	Requires minimal coding knowledge.
Developers manually write most of the application code.	Applications are built using visual drag-and-drop components.
Development generally takes more time.	Applications can be developed much faster.
UI, database and workflows often need separate development.	Forms, reports, databases and workflows can be configured in one platform.
Testing and debugging require more development effort.	Many validations and features are available as built-in components.
Changes usually require code modification.	Changes can often be made through visual configuration.
Suitable for highly customized and complex applications.	Suitable for business applications and rapid application development.
Maintenance usually requires dedicated developers.	Business users and developers can collaboratively maintain applications.

Emerging Trends in Low-Code Development

- Drag-and-drop application builders
- Visual workflow automation
- Cloud-based application development
- AI-assisted application development
- Integration with APIs and external services
- Reusable components and templates
- Rapid development and deployment
- Mobile-friendly application development
- Low-code database and report generation
- Citizen development, where non-professional programmers can build applications.

3. Benefits and Challenges in Low-Code

Benefits

1. Faster Development - Applications can be developed quickly using ready-made visual components.
2. Less Coding - Reduces the amount of traditional programming required.
3. Easy to Use - Drag-and-drop interfaces make development easier for beginners.
4. Easy Maintenance - Forms, fields and workflows can be modified with less effort.
5. Automation - Notifications, emails, approvals and workflows can be automated.
6. Reduced Development Cost - Shorter development time and reusable components can reduce overall cost.
7. Rapid Prototyping - Applications can be tested, demonstrated and modified quickly.
8. Cloud Accessibility - Applications can be accessed through the internet from different devices.

Challenges

1. Platform Dependency - Applications may become dependent on a particular low-code platform.
2. Limited Customization - Highly complex requirements may still require additional coding.
3. Security Concerns - Applications must be configured properly to protect user and business data.
4. Scalability Issues - Very large or complex applications may require additional resources and careful design.
5. Integration Limitations - Some external systems may require custom APIs or custom code.
6. Vendor Lock-in - Moving an application to another platform can be difficult.
7. Learning Platform-Specific Features - Users still need to understand the platform tools, rules and limitations.

4. Comparison of Existing Low-Code Platforms

Feature	Zoho Creator	Microsoft Power Apps	Google AppSheet	Mendix
Development Style	Low-code / visual	Low-code / visual	No-code / low-code	Low-code
Form Builder	Yes	Yes	Yes	Yes
Database Support	Built-in	Dataverse	Built-in / external sources	Built-in
Workflow Automation	Yes	Power Automate	Automation Bots	Workflow / Microflows
Reports & Dashboards	Yes	Yes	Yes	Yes
Mobile Application	Yes	Yes	Yes	Yes
AI Features	Available	Available	Available	Available
API Integration	Yes	Yes	Yes	Yes
Suitable For	Business applications	Enterprise applications	Data-driven applications	Enterprise applications
Main Advantage	Easy application and workflow development	Strong Microsoft ecosystem integration	Simple and fast app creation	Enterprise-level application development

Application Allocation

Roll No.: 240701187 Ending digit: 7

Therefore, the application to be used for the practical implementation is the 7th application from the application list posted on 25 August.

5. RESULT

The survey on emerging trends in low-code application development was successfully completed. The study showed that low-code platforms use visual development, reusable components, workflow automation, cloud services and AI-assisted features to reduce development effort and speed up application delivery.

The comparison of Zoho Creator, Microsoft Power Apps, Google AppSheet and Mendix shows that all four platforms support visual application development, workflow automation, mobile access, dashboards and API integration. Each platform differs mainly in its ecosystem, database support, level of customization and intended users.

Hence, low-code development is useful for rapid prototyping and for building many real-world business applications with less traditional coding, while challenges such as vendor dependency, security, scalability and customization must also be considered.